Preface

Kubernetes would like to thank every sysadmin who has woken up at 3 a.m. to restart a process. Every developer who pushed code to production only to find that it didn't run like it did on their laptop. Every systems architect who mistakenly pointed a load test at the production server because of a leftover hostname that they hadn't updated. It was the pain, the weird hours, and the weird errors that inspired the development of Kubernetes. In a single sentence: Kubernetes intends to radically simplify the task of building, deploying, and maintaining distributed systems. It has been inspired by decades of real-world experience building reliable systems, and it has been designed from the ground up to make that experience if not euphoric, at least pleasant. We hope you enjoy the book!

Who Should Read This Book

Whether you are new to distributed systems or have been deploying cloud native systems for years, containers and Kubernetes can help you achieve new levels of velocity, agility, reliability, and efficiency. This book describes the Kubernetes cluster orchestrator and how its tools and APIs can be used to improve the development, delivery, security, and maintenance of distributed applications. Though no previous experience with Kubernetes is assumed, to make maximal use of the book, you should be comfortable building and deploying server-based applications. Familiarity with concepts like load balancers and network storage will be useful, though not required. Likewise, experience with Linux, Linux containers, and Docker, though not essential, will help you make the most of this book.

Why We Wrote This Book

We have been involved with Kubernetes since its very beginnings. It has been truly remarkable to watch it transform from a curiosity largely used in experiments to a crucial production-grade infrastructure that powers large-scale production applications in varied fields, from machine learning to online services. As this transition occurred, it became increasingly clear that a book that captured both how to use the core concepts in

Kubernetes and the motivations behind the development of those concepts would be an important contribution to the state of cloud native application development. We hope that in reading this book, you not only learn how to build reliable, scalable applications on top of Kubernetes but also receive insight into the core challenges of distributed systems that led to its development.

Why We Updated This Book

The Kubernetes ecosystem has continued to grow and evolve since the first and second editions of this book. There have been many Kubernetes releases, and many more tools and patterns for using Kubernetes have become de facto standards. In the third edition, we focused on the addition of topics that have grown in interest in the Kubernetes ecosystem including security, accessing Kubernetes from programming languages, as well as multicluster application deployments. We also updated all of the existing chapters to reflect the changes and evolution in Kubernetes since the first and second editions. We fully expect to revise this book again in a few years (and look forward to doing so) as Kubernetes continues to evolve.

A Word on Cloud Native Applications Today

From the first programming languages, to object-oriented programming, to the development of virtualization and cloud infrastructure, the history of computer science is a history of the development of abstractions that hide complexity and empower you to build ever more sophisticated applications. Despite this, the development of reliable, scalable applications is still dramatically more challenging than it ought to be. In recent years, containers and container orchestration APIs like Kubernetes have proven to be an important abstraction that radically simplifies the development of reliable, scalable distributed systems. Containers and orchestrators enable developers to build and deploy applications with a speed, agility, and reliability that would have seemed like science fiction only a few years ago.

Navigating This Book

This book is organized as follows. Chapter 1 outlines the high-level benefits of Kubernetes without diving too deeply into the details. If you are new to Kubernetes, this is a great place to start to understand why you should read the rest of the book.

Chapter 2 provides a detailed introduction to containers and containerized application development. If you've never really played around with Docker before, this chapter will be a useful introduction. If you are already a Docker expert, it will likely be mostly review.

Chapter 3 covers how to deploy Kubernetes. While most of this book focuses on how to *use* Kubernetes, you need to get a cluster up and running before you start using it. Although running a cluster for production is outside the scope of this book, this chapter presents a couple of easy ways to create a cluster so that you can understand how to use Kubernetes. Chapter 4 covers a selection of common commands used to interact with a Kubernetes cluster.

Starting with Chapter 5, we dive into the details of deploying an application using Kubernetes. We cover Pods (Chapter 5), labels and annotations (Chapter 6), services (Chapter 7), Ingress (Chapter 8), and ReplicaSets (Chapter 9). These form the core basics of what you need to deploy your service in Kubernetes. We then cover deployments (Chapter 10), which tie together the life cycle of a complete application.

After those chapters, we cover some more specialized objects in Kubernetes: DaemonSets (Chapter 11), Jobs (Chapter 12), and ConfigMaps and Secrets (Chapter 13). While these chapters are essential for many production applications, if you are just learning Kubernetes, you can skip them and return to them later, after you gain more experience and expertise.

Next we introduce role-based access control (Chapter 14) and cover service meshes (Chapter 15) and integrating storage (Chapter 16) into Kubernetes. We discuss extending Kubernetes (Chapter 17) and accessing Kubernetes from programming languages (Chapter 18). We then focus on securing Pods (Chapter 19) along with Kubernetes policy and governance (Chapter 20).

Finally, we conclude with some examples of how to develop and deploy multicluster applications (Chapter 21) and a discussion of how to organize your applications in source control (Chapter 22).

Online Resources

You will want to install Docker. You likely will also want to familiarize yourself with the Docker documentation if you have not already done so.

Likewise, you will want to install the `kubectl` command-line tool. You may also want to join the Kubernetes Slack channel, where you will find a large community of users who are willing to talk and answer questions at nearly any hour of the day.

Finally, as you grow more advanced, you may want to engage with the open source Kubernetes repository on GitHub.

Conventions Used in This Book

The following typographical conventions are used in this book:

Italic

Indicates new terms, URLs, email addresses, filenames, and file extensions.

`*Constant width*`

Used for program listings, as well as within paragraphs to refer to program elements such as variable or function names, databases, data types, environment variables, statements, and keywords.

`Constant width bold`

Shows commands or other text that should be typed literally by the user.

`*Constant width italic*`

Shows text that should be replaced with user-supplied values or by values determined by context.

NOTE

This icon signifies a tip, suggestion, or general note.

WARNING

This icon indicates a warning or caution.

Using Code Examples

Supplemental material (code examples, exercises, etc.) is available for download at *https://github.com/kubernetes-up-and-running/examples*.

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Acknowledgments

We would like to acknowledge everyone who helped us develop this book. This includes our editors, Virginia Wilson and Sarah Grey, and all of the great folks at O'Reilly, as well as the technical reviewers who provided tremendous feedback that significantly improved the book. Finally, we would like to thank all of our first edition and second edition readers who took the time to report errata that were found and fixed in this third edition. Thank you all! We're very grateful.

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